

Intellectual property law & data(base) protection

TILBURG UNIVERSITY

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Understanding Society

Focus and Structure

Q. Does EU IP law allow data mining of **non-personal data** contained in a database that has been **made available to the public**?

I. KC Recap: What is IP law? Can data be protected under IP law?

II. Can databases be protected under IP law?

- Protection offered by *EU Database Directive (1996).*

III. Is it possible to engage in text and data-mining (DM) of a database protected by IP law?

- Exceptions provided in Articles 3 and 4 of the *EU Copyright in the Digital Single Market Directive (2019).*

IV. Contract law as an additional barrier to DM.

What is non-personal publicly accessible data?

- Non-personal (anonymized) data: Data that does not fall within the scope of the definition of personal data under Article 4(1) of EU GDPR.
  - Data that does not enable the identification of a particular data-subject.

- Publicly accessible databases

Databases that have been disclosed to the public in the sense that they have been published (online/off-line) and are available for consultation and use by members of the public even though use and access may be subject to legal (e.g. IP rights, contractual provisions)/ non-legal (digital rights management) restrictions.

I. KC Recap

Which of the following can be protected under IP law? Why?

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Table: customers

customer_id	first_name	last_name	phone	country
1	John	Doe	817-646-8833	USA
2	Robert	Luna	412-862-0502	USA
3	David	Robinson	208-340-7906	UK
4	John	Reinhardt	307-242-6285	UK
5	Betty	Taylor	806-749-2958	UAE



3.6 kg



Only products of (human?) intellect can be protected by IP law

KC: Can data be protected under IP law?

Data

A collection of raw (unrefined), unorganized symbols (e.g. numbers or text) that do not carry specific meaning unless processed, organized and structured within a specific context.

Q. Is data a product of human intellect?

- No. Individual units of raw-data are not protected under IP law *as such* because they do not typically qualify as products of human intellect.

Information

Data that is processed, organized and structured within a specific context in order to enable analysis and the obtaining of knowledge.

- In certain situations – information & informational goods can be protected under IP rights by virtue of the human intellectual effort invested in their production.

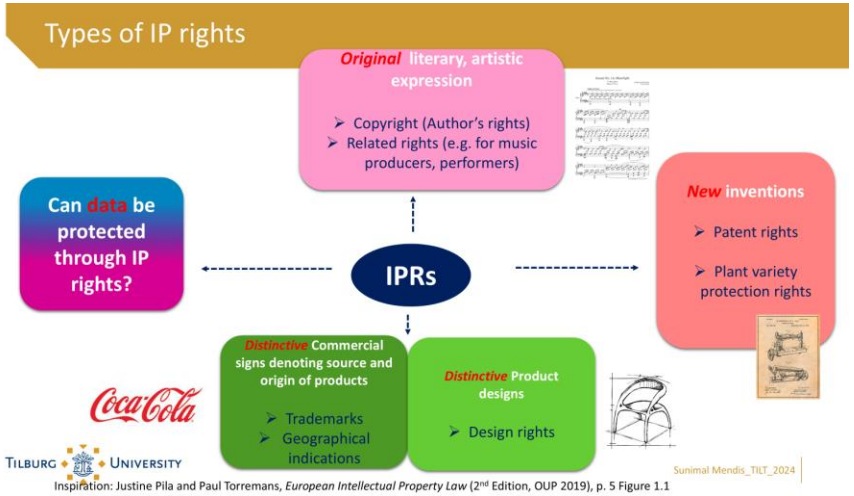
  - Information: Trade secrets [EU Trade Secrets Directive (2016)].

  - Collections of data: Databases [EU Database Directive (1996)].
    - Original databases: Copyright.
    - Non-original databases: *Sui generis* database rights.

If a piece of data/database is protected by IP law, what does that mean?

Certain specific acts (e.g, reproduction, adaptation, communication to the public, distribution (sale)) cannot be carried out without the rigththolders' authorization, unless it is permitted under an exception to the right.

The specific acts that cannot be carried out without authorization and the applicable exceptions can vary according to the type of IP right.



Inspiration: Justine Pila and Paul Torremans, *European Intellectual Property Law* (2nd Edition, OUP 2019), p. 5 Figure 1.1

Analyzing legal rights

Subject-matter	*What* is protected?
Criteria	*Which* criteria (conditions) need to be fulfilled for the subject-matter to obtain protection?
Scope	*How* is the subject-matter protected (i.e. against what kind of acts is it protected)?
Ownership	*Who* is the owner of the right (who is the rightholder)?
Duration	*For how long* is the subject-matter protected (what is the term of protection)?
Exceptions	*What* acts can be carried out without the rightholder's permission?

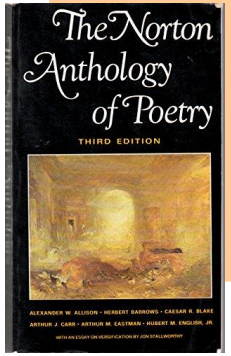
II. Database Protection

DD grants protection to databases in two ways (two regimes of protection)

Original databases

Copyright protection
Articles 3-6 of DD

Object: Protect human creativity expended in creating a database.



Non-original databases

Protection under *sui generis* database right
Articles 7-11 of DD

Object: Protect non-creative investment (financial, know-how, skill and labour, time) expended in creating a database.

S&P Global
Market Intelligence

A database could obtain protection under copyright *AND* the sui generis database right, at the same time if the criteria necessary for obtaining protection under each regime is fulfilled.

Subject-matter

What is a database?

A collection of independent **data** arranged in a **systematic or methodical way** and **individually accessible** by electronic or other means. [Article 1(2) DD]

Important:

- Protection **does not** extend to the individual units of data contained in the database (but such content may on their own qualify for protection e.g. the individual photographs incorporated in a photography catalogue).



Subject-matter

1. **Collection**: A collection or compilation of data.
   - Data could be in the form of text, sound, images, numbers, facts etc.

2. **Independence of data:** Data must constitute separate and independent items.
   - *Test:* Item(s) of data must have independent informative value when separated from database.

3. **Systematic/methodical arrangement:** Data must be arranged in a systematic or methodical way (arrangement is logical not random).

4. **Individual accessibility**: Independent units of data can be retrieved individually.
   - Collection should include technical or other means (e.g. electro-optical processes, an index or a particular plan or method of classification) to allow retrieval of independent items of data.

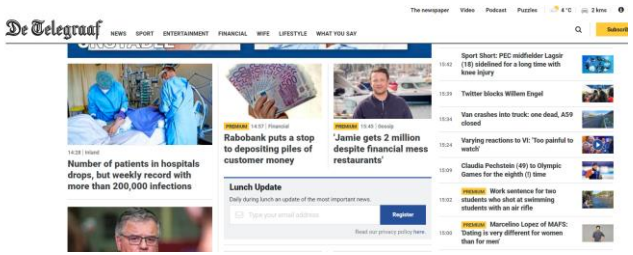
Key Features of a database

- ✓ Collection or compilation of data.
- ✓ Data constitute separate and independent items (each item of data can have independent informative value on its own).
- ✓ Data is arranged in a systematic or methodical way (arrangement is logical not random).
- ✓ Independent items of data can be retrieved individually (independently accessible).



Quiz 1: Database or not?

E-version of *De Telegraaf* newspaper



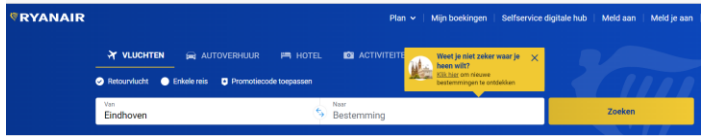
The movie “The Matrix” (as when you watch it in the cinema)



A map of Den Bosch as displayed on Google maps.

Your Facebook page

Data stored on the Ryanair website



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Copyright protection

Copyright protection

Subject-matter

Literary, artistic, scientific **expression** ("work") that is the product of human (?) intellect.

- A work is a creative combination of elements such as words, sounds, shapes or colors.

Criteria

Sole criterion: Originality (expression must be its **A**uthor's **O**wn **I**ntellectual **C**reation (***AOIC*** standard of originality).

Work must reflect the author's **personal creative choices** leading to his **personal touch being stamped** on the expression.



1. **Creator has had the freedom to engage in his own (arbitrary) creative decision-making.**
   - AOIC standard will **not** be fulfilled when the creation process is dictated by technical or functional considerations, rules or restrictions thereby leaving no space for the exercise of free and creative choices.

2. Creator has used this freedom to make personal creative choices in creating the expression. By reason of the creator making his personal creative choices the expression is stamped with the creator's personal touch.

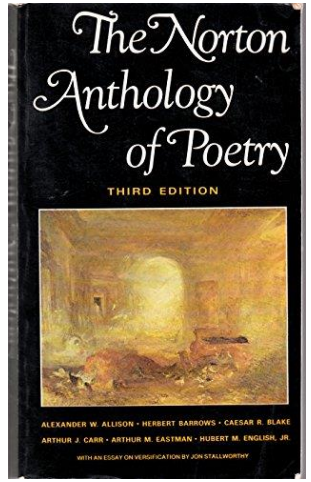
   - Mere expenditure of **skill, labour and know-how** is **insufficient** for fulling AOIC standard of originality!

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Copyright protection for databases

Art. 3(1) DD, Sole criterion: **Originality**

Database must be original → (***AOIC*** standard of originality).
The selection **AND/OR** arrangement of the contents of the constitute its the **A**uthor's **O**wn **I**ntellectual **C**reation



Criteria: Checklist

Does the database fulfill the AOIC standard of originality in the selection OR arrangement of contents?

Selection of contents in database

- ✓ Creator has had freedom to make his personal creative choices in selecting contents.

  Yes ⬇ No 

- ✓ Creator has used this freedom to make his personal creative choices in selecting contents.

  Yes ⬇ No 

- ✓ By reason of the creator making his personal creative choices the selection of the contents of the database is stamped with the creator's personal touch.

  Yes  No 

Arrangement of contents in database

- ✓ Creator has had freedom to make his personal creative choices in arranging contents.

  Yes ⬇ No 

- ✓ Creator has used this freedom to make his personal creative choices in arranging contents.

  Yes ⬇ No 

- ✓ By reason of the creator making his personal creative choices the arrangement of the contents of the database is stamped with the creator's personal touch.

  Yes  No 



Activity 2: Can the following obtain copyright protection?

(Refer to Guide for Solving Problem Questions on Canvas)

Answer

Step 1: Is it a database within the definition in Article 1(2) of DD?

➢ Definition : A collection of independent works, data or other materials arranged in a systematic or methodical way and individually accessible by electronic or other means.

Are all the following criteria followed?

- Is it a collection or compilation of data? Yes/No/Uncertain? Give reasons.
- Does each item of data have independent informative value on its own? Yes/No/Uncertain? Give reasons.
- Is data is arranged in a systematic or methodical way? Yes/No/Uncertain? Give reasons.
- Can independent items of data be retrieved individually? Yes/No/Uncertain? Give reasons.

Step 2: If answer to Step 1 is yes, can database obtain protection under copyright?

➢ Test in Article 3(1) of DD: Does selection/arrangement of database fulfill AOIC standard of originality by qualifying as its author's own intellectual creation?

Does the selection and/or arrangement of the database fulfill the AOIC standard of originality? Yes/No/Uncertain? Give reasons.

A. Louvre Museum collection

The Louvre Museum in France has a large collection of renaissance portrait paintings that were created during the 15th-16th Centuries. Vincent is a curator at the Louvre Museum.

- Vincent evaluates over 300 digital images of renaissance portrait paintings stored in the Museum's digital archives and selects 100 digital images which, according to his own creative sense represent the most beautiful renaissance portrait paintings held by the Museum.
- He categorizes and arranges these 100 digital images according to five different themes (i.e., “contemplation”, “innocence”, “bliss”, “sorrow”, “pride”) based on the emotions that, in his opinion, are evoked by each painting.
- The sequencing of paintings within each theme is also done according to Vincent’s personal creative preferences.

The thematically arranged images are published on the Louvre Museum Website as a digital database entitled the “100 most beautiful renaissance portrait paintings”.

B. Simson's Database

- **Simson** is a Dutch company registered in the Netherlands. Its registered office and principal place of business is in Eindhoven.
- Simson spends 20 million EUR per year to purchase viewership data (i.e., data on audience sizes) for TV-programs broadcast in different EU Member States from various media research companies. Simson spends an extra 5 million EUR per year to check the accuracy of the data it purchases.
- Simson's employees expend much skill and labor in systematically arranging this data according to the language of the TV program to form a database (e.g., the data is categorized as English language TV-programs, Spanish language TV-programs, Dutch language TV-programs etc.).

Ownership

Art. 4

- Author: Natural person who carried out selection and arrangement of contents, OR
- Employer of that natural person, which could be a natural or legal person (e.g., company).

Duration

Database protected during the lifetime of author (natural person who carried out selection and arrangement of contents) + 70 years.

Scope

Art. 5 DD: Engaging in following acts (for commercial/ non-commercial purposes) require copyright owner's authorization:

a) Temporary/ permanent **reproduction** (copying) of database in whole or in part, by any means and in any form (directly/indirectly). [Does not cover mere consultation of a database made available to the public].

b) **Distribution** (sale) of database or its copies to the public.

c) **Adaptation** or modification of database (e.g. translation).

d) **Communication**, display or performance of the database **to the public**.

[Communication: transmission of the database by wire/wireless means including the making available to the public in such a way that members of the public may access them from a time and place individually chosen by them i.e., interactive on-demand transmissions].

- Communication should be made to a "new public", i.e., a public to whom the rightholder did not intend to make work available. E.g., A made copyright protected database available only to licensed subscribers. B copied the database without A's permission and posted it on his personal website, thereby making it available also to those who had not obtained a license. These persons constitute a "new public".

Infringement test: Copyright.

- ✓ Database is protected by copyright (AOIC standard of originality is fulfilled; protection has not expired).

  Yes → 

  No → 

- ✓ The intended acts come within the scope of copyright in the database.

  Yes → 

  No → 

- ✓ You have authorization of the copyright owner to carry out the intended acts (e.g., license) OR the intended acts fall within a legally granted exception to copyright in the database.

  Yes → 

  No → 

Ownership

Art. 4

- Author: Natural person(s) who carried out selection and arrangement of contents, OR
- Employer of that natural person(s).

Duration

EU Copyright Term Directive **Art. 1:**

Database protected during the lifetime of author (natural person who carried out selection and arrangement of contents) + 70 years.

Louvre Museum collection

The “100 most beautiful renaissance portrait paintings” digital database is made available free of charge on the Louvre Museum’s website, but only to those who have registered themselves as users of the Louvre Museum website.

Jan is an Art Historian living in Amsterdam.

- Jan copies 50 of the “100 most beautiful renaissance portrait paintings” onto his own computer and arranges them in the same way they were arranged by Vincent.

  Direct reproduction of a part: Reproduction right

- He prints them as a booklet entitled “50 renaissance Masterpieces” which he sells to the public at 25 euros a booklet. The booklet includes information on each painting and his own critical commentary on why each painting constitutes a masterpiece.

  Adaptation right + Distribution right

An electronic low-resolution version of the booklet is made available on his personal website free-of-charge, so that any member of the public can access his booklet.

Communication to the public right [new public]

Sui generis protection

Preliminary condition: Who can benefit from protection?

Article 11 DD: ***Sui generis right can only be obtained if the maker of the database is:***

- Natural person: Who holds nationality of an EU Member State or maintains habitual residence in the territory of the EU.

- Legal person: A company or firm that fulfills ***both*** of following criteria.

  1. Formed in accordance with the law of an EU Member State and having its registered office within the community.

  AND;

  2. Having its central administration **or** principal place of business within the Community. [Because operations of company must be genuinely linked on an ongoing basis with the economy of an EU Member State].



Activity 3: Can the following persons/entities potentially benefit from the *sui generis* database right?

Can the following persons/entities potentially benefit from the *sui generis* database right?

- Simson is a Dutch company registered in the Netherlands. Its registered office and principal place of business is in Eindhoven.
- Yin-Shi is registered as a company in China. It maintains its registered office in Beijing but has corporate offices in Amsterdam, Paris and Berlin.
- Vincent is a French citizen who maintains his habitual place of residence in Amsterdam.
- Sheila is a Nigerian national. She is registered as a student at the University of Tilburg but follows her lectures over Zoom from Lagos, Nigeria.

Criteria

Art. 7 DD: There must have been a substantial investment made in either the obtaining, verification or presentation of contents of database.

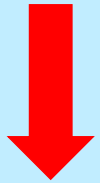
- Obtaining: investment made in seeking out and collecting existing data (even if the initial objective of collecting the data was not for creating the database per se.
- Verification: investment made in ensuring the accuracy/reliability of the data during collection and afterwards (periodic monitoring the 'materials' of a database to ensure completeness and accuracy).
- Presentation: Selection and/or arrangement of contents in a way that is individually accessible.

Investment: Can be financial, human effort, expertise, time, know-how, skill and labour etc.

Criteria: Checklist

I. Preliminary condition: Can creator benefit from protection under Article 11 DD?

Yes



No



II. Has there been a substantial investment in **<u>one or more</u>** of the following as per Article 7 DD?

Obtaining contents?

Yes 

No 

Verification of contents?

Yes 

No 

Presentation of contents?

Yes 

No 



Activity 4: Can the following obtain sui generis database rights protection?

Steps to reaching your answer

Step 1:Is it a database within the definition in Article 1(2) of DD?

Step 2: If answer to Step 1 is yes, are preliminary conditions for obtaining protection (Article 11 of DD) fulfilled?

- Natural person: EU citizen or maintains habitual residence within EU? Yes/No/Uncertain? Give reasons for your answer.
- Legal person: Company registered in EU with registered office in EU + principal place of business or central administration in territory of EU? Yes/No/Uncertain? Give reasons for your answer.

Step 3: If answer to Step 2 is yes, are the criteria for obtaining sui generis protection (Article 7 DD) fulfilled? Yes/No/Uncertain? Give reasons for your answer.

- Has an investment in either the obtaining, verification or presentation of contents of database? Yes/No/Uncertain? Give reasons for your answer.
- Is this investment substantial? Yes/No/Uncertain? Give reasons for your answer.

Simson's Database

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- Simson spends 20 million EUR per year to purchase viewership data (i.e., data on audience sizes) for TV-programs broadcast in different EU Member States from various media research companies. Simson spends an extra 5 million EUR per year to check the accuracy of the data it purchases.
- Simson's employees expend much skill and labor in systematically arranging this data according to the language of the TV program to form a database (e.g., the data is categorized as English language TV-programs, Spanish language TV-programs, Dutch language TV-programs etc.).

Difference between creation and obtaining of data

Investment made in creating (generating) data or verifying data at the point of creation is not relevant in determining whether a substantial investment has been made in obtaining or verifying contents of database!

“Spin-off” databases (i.e., compilations of data that are generated as by-products ('spin-offs') of other activities cannot claim sui generis database protection on the basis of having made an investment in the obtaining (i.e., collection) of data, because the investment is not made in obtaining existing data but in engaging in a different activity as a result of which the data is generated.

E.g., A TV program schedule that is generated by a TV station, a database of flight schedules generated by an airline company.

[more information, https://www.ivir.nl/publicaties/download/spinofffordham.pdf]



Activity 4: Is the investment in the creation or obtaining of data?

British Horseracing Board (BHB) database [Case C-203/02, *BHB v William Hill* (2004)]

BHB organizes horse-races in the UK. In order to, organize these races it registers information concerning owners, trainers, jockeys and horses, decides on the horses to be entered for each race (based on their weight, handicapping etc.) and finally it compiles the lists of horses running in the races. This requires a large financial investment (approx. £4 million per annum).

The data is entered into the BHB database.

N.B. This case took place before Brexit, so BHB fulfills the preliminary condition under Article 11 DD.

Answer

Investments related to the creation of data and verifying the accuracy of the data at the point of creation but not to the creation of the database (i.e. not to the selection or arrangement of the data)!!

But,

Observed data (e.g. compilation of data about football matches in progress including goals and goalscorers, yellow and red cards and which players were given them) is obtained data, not created data.

[Football Dataco & Others v Stan James [2013] England & Wales Court of Appeal Civ 27].

Scope

Art. 7: Engaging in following acts (for commercial/ non-commercial purposes) require sui generis right owner's authorization:

- Extraction and/or re-utilization of whole or substantial part of contents of database evaluated qualitatively and/ or quantitatively.
  - *Includes,* repeated and systematic extraction and/or re-utilization of insubstantial parts of the contents of database where this is **contrary to the normal exploitation of that database or unreasonably prejudices the legitimate interests of the producer of the database.** [Art. 7(5) DD].

How to test "Substantiality

- "Quantitative: Volume of data assessed in relation to total volume of contents of database.
- Qualitative: Scale of investment in obtaining, verification and presentation of contents extracted/re-utilized is substantial.

Scope

- **Extraction** ⟷ Reproduction (copying)

Appropriation and transfer of contents of database from one medium to another (i.e. direct/indirect reproduction). (CJEU, C-203/02, *BHB v William Hill* [2004]).

Does not require physical copying. Transferring the elements of one database to another database following on-screen visual consultation of the first database will constitute extraction. (CJEU, C-304/07, *Directmedia Publishing GmbH v Albert-Ludwigs-Universität Freiburg* [2008]).

- **Re-utilization** ⟷ Any form of making available contents (either in their original form or in modified form) of a database to the public by distribution (sale); communication, display or performance (open-ended).

Infringement test: Sui generis database right.

- ✓ Database is protected by the sui generis database right (preliminary condition + criteria for protection fulfilled; protection has not expired).

  Yes → 

  No → 

- ✓ The intended acts come within the scope of the sui generis database right.

  Yes → 

  No → 

- ✓ You have authorization of the sui generis database rights holder to carry out the intended acts (e.g., license) OR the intended acts fall within a legally granted exception to the sui generis database right.

  Yes → 

  No → 

- ✓ You extract/re-utilize a substantial part of the contents of the database OR You extract/re-utilize repeatedly and systematically insubstantial parts of the contents of database in a way that is contrary to the normal exploitation of that database or unreasonably prejudices the legitimate interests of the producer of the database.

  Yes → 

  No → 



Activity 6: Does the unauthorized act come within the scope of the sui generis database right? Is there *potential* infringement?

(Guide for Solving Problem Questions on Canvas, Steps 1-6)

“Romanian-language TV programs” is the smallest category of data in Simson’s database. It comprises less than 10% of the total volume of the database. However, obtaining and verifying the data in this category was very expensive and accounts for more than 35% of the total investment made in the creation of the database.

Jan has access to Simson’s database. Without Simson’s authorization, Jan copies all the data included in the category of “Romanian-language TV programs” onto his own hard-drive.

Does Jan’s act infringe on the sui generis database right?

a. Yes. Because he has extracted a substantial part of the contents of the database.

b. Yes. Because he has re-utilized a substantial part of the database.

c. No. Because he has only extracted an insubstantial part of the database.

d. No. Because he has only re-utilized an insubstantial part of the database.

Duration

Art. 10 DD:
Database protected for 15 years, calculated from 1 January of the year *after* date of completion of database.

- **Art. 10(3) DD**: 'rolling' *sui generis* right: Any 'substantial change' (i.e., additions, deletions, re-arrangement, verifications to the contents of a database) that requires a substantial new investment evaluated qualitatively or quantitatively shall qualify the database as a whole to enjoy a new period of protection.

Ownership

Art. 7 DD: Natural person or legal person who is responsible for the substantial investment made in the creation of the database.

The curious case of machine-generated databases

Machine-generated databases (MGDs)

Databases that are created, updated, and managed primarily by automated processes and machines (including the IoT).

Some Characteristics:

- Dynamic: Data is continuously or periodically collected, modified and/or updated (high velocity, real-time). The structure of the database can also be subject to change.

- Can machine-generated dynamic databases obtain copyright protection?

- Can machine-generated dynamic databases obtain sui generis protection?

Substantial investment?

Difficult to distinguish between 'creating' and 'obtaining' when big data is generated, aggregated, processed and exchanged in real-time.

- Gervais: Big Data corpora are neither "obtained" nor "collected"; they are generated by the machine. [Gervais, 2019, 29].

- But, sometimes machine-generated databases generate data based on observed occurrences.
  - ➢ Data from an automatically administered database can be deemed as 'obtained' when it records a physical or digital, objectively appreciable occurrence (i.e., data observed and recorded by a satellite).

Leistner: Investment into the establishment of an infrastructure for obtaining, verifying or presenting pre-existing data could be relevant for assessing substantiality of the investment

➢ Could substantial investments in developing a technological (e.g., AI-based) infrastructure for aggregating, cleaning and organizing the data in a machine-generated dynamic database be considered a substantial investment in the verification and presentation of data?

Machine-generated databases excluded from protection under the sui generis database right

CHAPTER X

SUI GENERIS RIGHT UNDER DIRECTIVE 96/9/EC

Article 35

Databases containing certain data

The *sui generis* right provided for in Article 7 of Directive 96/9/EC shall not apply when data is obtained from or generated by a product or related service falling within the scope of this Regulation, in particular in relation to Articles 4 and 5.

Recital (84) In order to eliminate the risk that holders of data in databases obtained or generated by means of physical components, such as sensors, of a connected product and a related service or other machine-generated data, claim the sui generis right under Article 7 of Directive 96/9/EC, and in so doing hinder in particular the effective exercise of the right of users to access and use data and the right to share data with third parties under this Regulation, it should be clarified that the sui generis right does not apply to such databases. That does not affect the possible application of the *sui generis* right under Article 7 of Directive 96/9/EC to databases containing data falling outside the scope of this Regulation provided the requirements for protection in accordance with Article 7(1) of that Directive are fulfilled.

III. Exceptions for data mining (DM)

Definition of data mining

Art. 2(2) EU Copyright in the Digital Single Market (DSM) **Directive (2019)**:

Any automated analytical technique aimed at analyzing text and data in digital form in order to generate information which includes but is not limited to patterns, trends and correlations.

For purpose of this course,
“data mining” = “text and data mining”

Legal barriers to data mining

a) Copyright (or other IP rights) in the data.

b) Trade secrecy in the data and/or database.

c) Copyright/sui generis database right in the database.

d) Contract law: Contractual terms and conditions that prevent data-mining either of database and/or of individual data.

Scope of protection

Copyright

Art. 5

a) Temporary/ permanent reproduction of database in whole or in part, by any means and in any form (directly/indirectly).

a) Distribution of database (or copies) to the public.

b) Adaptation or modification of database.

c) Communication, display or performance of the database to the public.

Sui generis right

Art. 7

Extraction and/or re-utilization of whole or substantial part of contents of database evaluated qualitatively and/ or quantitatively.

- Extraction: Temporary/ permanent reproduction of database in whole or in part, by any means and in any form (directly/indirectly).
- Re-utilization: Distribution; adaptation or modification; communication, display or performance to the public.

Includes, repeated and systematic extraction and/or re-utilization of **insubstantial parts** of the contents of database where this is contrary to the normal exploitation of that database or **unreasonably prejudices the legitimate interests of the producer of the database.**

Database protection as a legal barrier to data-mining

Copying and loading data into a dataset (e.g. converting them into machine readable format and uploading data to own server or platform.)

Analysis of dataset by running the algorithm to identify and evaluate correlations between the data.

Recombination (rearrangement) of the data to obtain and visualize results.

Storage and communication of results to peers/public (e.g. publication of dataset).

Copyright: Temporary/ permanent reproduction (direct/indirect) of database in whole or in part.

Sui generis right: Extraction of whole or substantial part of contents of database.

Copyright: Distribution/Communication of database to the public.

Sui generis right: Re-utilization of whole or substantial part of contents of database.

Exceptions

What acts can be carried out without the rightholder's permission?

Exceptions for text-and-data mining (TDM) provided under Arts. 3 & 4 of the EU Copyright in the Digital Single Market (DSM) Directive (2019).

Article 3 DSM: Mandatory exception permitting data-mining for purposes of scientific research

Scope	✓ **Reproductions** and **extractions** of **lawfully accessible** databases protected by copyright and/or sui generis database right for **purposes of scientific research** (commercial + non-commercial). ✓ **Storage** of above reproductions and extractions with **appropriate level of security for purposes of scientific research (including for the verification of research results)**.
Beneficiaries	**Art. 2(1)**: **Research organizations** and **cultural heritage institutions** that have **lawful access** to the database and fulfill the following criteria: ✓ Primary goal is to **conduct scientific research**, OR to conduct both educational activities & scientific research; ✓ On a **non-profit basis** OR pursuant to a **public interest mission** (even if it is a for-profit entity). ✓ For-profit (commercial) undertaking will not have preferential access to and use of the results generated by the scientific research. ➢ *But,* mere fact that research is carried out in partnership with a for-profit entity will not prevent research organization or cultural heritage institution from benefitting from exception *if* for-profit entity will not obtain preferential access to and use of results. (Recitals 11&12).
Can it be overridden by contract?	**No**. **Art. 7**: Any contractual provision contrary to Art. 3 shall be unenforceable.

What is scientific research under Article 3 DSM?

Indication provided in Recital 12 DSM,

The term 'scientific research' within the meaning of this Directive should be **understood to cover both the natural sciences and the human sciences.** […]. Despite different legal forms and structures, research organisations in the Member States generally have in common that they act either on a not-for-profit basis or in the context of a public-interest mission recognised by the State. Such a public-interest mission could, for example, be reflected through public funding or through provisions in national laws or public contracts.

Conversely, organisations upon which commercial undertakings have a decisive influence allowing such undertakings to exercise control because of structural situations, such as through their quality of shareholder or member, which could result in preferential access to the results of the research, should not be considered research organisations for the purposes of this Directive.

Article 3: Check-list

Is DM undertaken for purposes of scientific research?

Yes

No



Is DM undertaken by an entity whose primary goal is to engage in scientific research on a non-profit basis or pursuant to a public interest mission?

Yes

No



Does entity have lawful access to the database?

Yes

No



Will a commercial entity have preferential access to research results?



Yes

No



Article 4 DSM: Mandatory exception permitting data-mining for any purpose

Scope	✓ **Reproductions and extractions of lawfully accessible databases protected by copyright and/or sui generis database right.** ✓ **Retention of above reproductions and extractions for the duration necessary for purposes of data mining.** ➢ No long-term storage possible as under Art. 3 exception!
Beneficiaries	**Any person** who **lawfully accesses** database.
Can it be overridden by contract?	**Yes**. According to Article 4(3) Owner of copyright/ sui generis database right can **opt-out** of Art. 4 exception by 'expressly reserving rights to data mining' But subject to following conditions (Recital 18): ✓ Publicly available database: opt-out should be signaled in **machine-readable format** (e.g. robot.txt metadata or sitemap) ✓ Not-publicly available database: sufficient signal opt-out in a contract or by unilateral declaration.

Article 4: Check-list

Does person engaging in datamining have lawful access to database?

Yes

No



Has rightholder opted-out of Art. 4 exception as set out in Recital 18?

No



Yes



Activity: Summarize key differences between Article 3 & 4 DSM

Article 3 DSM

- Scope of Application
  - Who? Limited to specific beneficiaries with lawful access.
  - What? Reproduction of database and copyright protected data included in database.
  - When? Data mining for purposes of scientific research, commercial & non-commercial.
    - Not available if for-profit entity will obtain preferential access to and use of results.
- Storage of data
  - Long-term storage of data possible for purposes of scientific research (with appropriate level of security).
- Contractual override
  - Not possible.

Article 4 DSM

- Scope of Application
  - Who? Open to anyone with lawful access.
  - What? Reproduction of database and copyright protected data included in database.
  - When? Data mining for any purpose, commercial & non-commercial.
- Storage of data
  - Only for as long as is necessary for purposes of DM (long-term storage of data not possible).
- Contractual override
  - Possible through opt-out exercised as stipulated in Article 4(3) DSM & Recital 18.

What is **lawful access**?

When the database is accessed via a lawful source [CJEU, C 435/12, *ACI Adam* [2014]; C-463/12 *Copydan* [2015]]

- Access of database is with rightholders' consent.

E.g. Accessing a database made available under an open access license or by an authorized person under a contract that permits such access (Recital 14 of DSM Directive).

If you have accessed content which has been made available *without* rightholders' consent (via an unlawful source), user must **not have known nor had reasonable grounds to know** that it was made available without rightholder's consent.

- But, such knowledge is presumed in the case of use for 'profit-making activities'.

What is **lawful use**? Who is a **lawful user**?

Cumulative criteria for assessing “lawful use”.

1. Access is via a lawful source - The content is made available to the public with the rightholder’s consent.
   - Lawful access is a precondition for lawful use, even if the use is covered by an exception or limitation [CJEU, C 435/12, *ACI Adam* [2014]].

2. Specific use is authorized by the rightholder (e.g., contractual terms OR open-access license authorizing use).
   - Lack of warnings, or contractual restrictions should not be taken as implicit authorization.

   OR;

   Specific use is not prevented by law. E.g.,
   - There are no IP rights, contractual terms that prevent use;
   - Specific use is permitted under a legally granted exception that cannot be overridden by contract).

Simson's database

- Research organizations can access Simson's databases by obtaining a contractual license subject to the payment of a monthly subscription. The terms of the contract states that, *"The database can only be consulted (on-screen consultation) for purposes of research. Users are not permitted to engage in datamining of the contents of this database"*.

- The Technical University of Eindhoven (TuE) has obtained a contractual license that enables its students and employees to consult Simson's database for their scientific research.

Ana is employed as a researcher at TuE.



She is working in a scientific research project that aims to develop an AI-based application that can predict the best type of entertainment programs to produce for different EU member states.

The research project is fully funded by a privately-owned (for-profit) market research company in Eindhoven called Spitz. Under the funding contract, once it is developed, the AI-application will be made available to the public free-of-charge and free-of-restrictions. Spitz is hoping to use the AI-application to boost its own profit-making activities.

For the purpose of training the application, Ana uses a software tool to mine the **entirety of Simson's database**. As soon as the data-mining is completed she immediately deletes all mined data.

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Joost is a data-science student at the Technical University of Eindhoven (TuE). He is employed as an intern at Spitz for the duration of the Summer vacation.

In order to prepare the market analysis requested by a Romanian client, Joost (without obtaining Simson's authorization) scrapes Simson's database for all the data on Romanian-language TV-programs. He stores the mined data securely on his hard-drive since he believes it can come in useful for his future research.

IV. Contract law as an additional barrier to DM of unprotected Databases.

Contract law



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Robots and Screen Scraping: You may not use data mining, robots, screen scraping, or similar data gathering and extraction tools on this site, except with our express written consent as noted below.

CJEU, *C-30/14, Ryanair Ltd v PR Aviation BV* [2015]

Ryanair is a low-cost airline company. It has been registered as a company in Ireland and its registered office and principal place of business is in Dublin.

The Ryanair website includes a database on flight data containing information on flights schedules, flight routes, and ticket-prices, to enable users to find the best flight options that match their queries and book them online.

Both the website and the database there are made available to members of the public subject to contractual terms including the following that any user to the website is required to accept.

> *You are not permitted to use this website other than for the following, private, non-commercial purposes: (i) viewing this website; (ii) making bookings (…)"*
>
> *'The use of automated systems or software to extract data from this for commercial purposes, ('screen scraping') is prohibited unless the third party has directly concluded a written licence agreement with Ryanair (…)'*

CJEU, *C-30/14, Ryanair Ltd v PR Aviation BV* [2015]

PR Aviation operates a website on which consumers can search through the flight data of low-cost air companies and compare prices. The data is obtained from the websites of individual airline companies (e.g. Ryanair) through an automated (screen-scraping) system.

Ryanair sued PR Aviation for breach of contract.

PR Aviation argued that they are permitted to engage in data mining under the exception to database protection provided in Art. 6(1) of DD which *cannot* be overridden by contract?

- Art. 6(1) of DD: Lawful user can engage in acts necessary for access to and normal use of the contents of a database.

CJEU, *C-30/14, Ryanair Ltd v PR Aviation BV* [2015]

CJEU said **NO!**

- Ryanair's database does NOT benefit from protection either under copyright OR the sui generis right.

- Article 6(1) exception is therefore not applicable.
  - If a database is not protected under the Database Directive (either copyright or sui generis right) the owner has the freedom to adopt of contractual clauses concerning the conditions of use of such a database.

So, exceptions and limitations or rights of lawful users that apply to databases protected under the Database Directive (including datamining exceptions under CDSM) **do not apply to unprotected databases!**